

Toward Symbolic-Aware Music Language Models

Compact Learned Symbolic Planning Tokens for Controlled Music Fusion

Dabin Kim Minhee Lee Ji Jiaxian Aner Zheng

Graduate School of Cultural Technology, KAIST

Central question: Does a compact learned symbolic stream, aligned to the audio window's own MIDI, reduce acoustic-token uncertainty more than a shuffled or dummy stream of the same length?

Audio Tokens Do Not Reliably Encode Musical Structure

What audio tokens can encode	What they miss
Timbre, loudness, texture	Key signature, chord progression
General instrument colour	Stem-level instrument identity
Rough tempo feel	Exact meter and beat phase
High-level scene	Phrase boundaries, section structure

MIDI encodes these exactly: note events, instrument tracks, tempo and key events. We learn a *compact* representation of this information and inject it into a LALM.

Hypothesis: an aligned symbolic stream reduces acoustic-token uncertainty — but only if it carries music-structural content, not just extra capacity. Shuffled and dummy controls are required at every step.

Diagnostic: Frozen UniAudio 2.0 Fails on Music QA

151 Slakh clips, 906 instrument-presence QA rows — no fine-tuning

Metric	Value
Exact accuracy	0.618
Macro-F1	0.378
Invalid responses (before recovery)	34.0%
Active invalid (after semantic recovery)	11.8%
Prediction distribution	796 yes / 3 no / 107 invalid

The model produces caption-like fragments instead of yes/no answers. Extreme yes-bias makes QA accuracy unreliable as a primary evaluation metric.

Implication: We rely on controlled cross-entropy comparisons and MIR probing — not open-ended QA accuracy — as our primary evaluation signals.

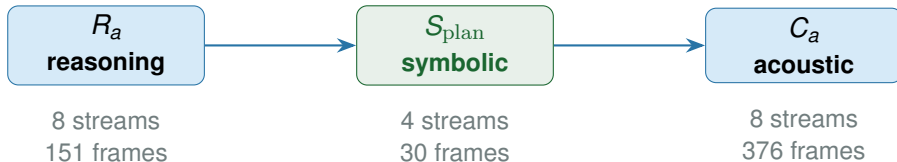
Base Model: UniAudio 2.0 Factorizes Audio into Two Streams

Stream	Role	Training
R_a — reasoning tokens	High-level, language-aligned	SFT + GRPO on ASR, captioning, reasoning
C_a — reconstruction tokens	Fine-grained acoustics (8 RVQ)	Trained after R_a frozen; 8 streams, 376 frames

A unified autoregressive transformer predicts C_a conditioned on text and R_a . Neither stream is supervised on MIDI-derived symbolic structure.

Gap: chord, meter, stem-level instrument identity are absent from both streams. We insert S_{plan} between R_a and C_a to carry exactly this missing information.

Our Approach: Insert S_{plan} as a Third Stream



Total sequence length: $151 + 30 + 376 = 557$ tokens — **0% truncation** under the 1,024 context limit.

Why 30 frames, not raw MIDI? 30 frames cover a full 30-second window (1 frame/second). Raw MIDI tokenization of the same window produces 2,000–8,000 tokens — context-budget-breaking and incompatible with RVQ positional encoding.

Evaluation Design: Four-Condition Control Protocol

Every experiment runs all four conditions simultaneously

Condition	Content of S_{plan} slot	What a gain here proves
Aligned S_{plan}	Tokens from the <i>same</i> window's MIDI	Symbolic content + temporal alignment
Shuffled S_{plan}	Tokens from a <i>different</i> window's MIDI	Token statistics intact; alignment broken
Dummy stream	Zero / random RVQ codes	Token-budget / capacity effect only
Reason-only	No S_{plan} slot at all	Base model without any symbolic stream

Claim rule: aligned must beat *both* shuffled and dummy to establish that alignment-specific symbolic content drives the gain — not just extra tokens.

Dataset: Slakh2100 — Paired Audio and MIDI

Split	Tracks	Windows	Notes
Train	1,289	11,279	Track-level split before windowing
Validation	270	2,334	Zero track overlap with train
Test	151	1,385	Held-out; never seen during training

Slakh provides professional-grade synthesized multi-track audio with exact aligned MIDI. Track-level splitting prevents musical phrases from appearing in both train and test.

All cross-entropy numbers in this presentation are on the held-out **test set** unless labelled “valid”. All conditions use the same audio-MIDI windows.

EXP024: The S_{plan} Codec Reconstructs Faithfully

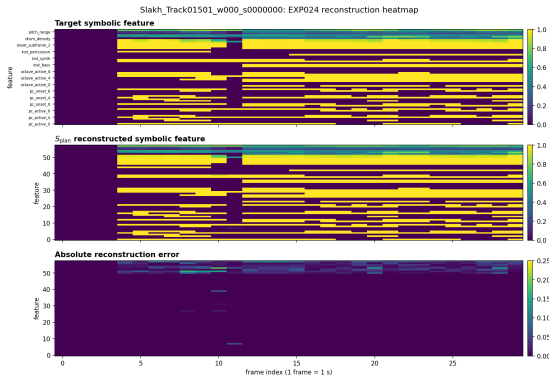
RVQ autoencoder — 4 streams, 512 codes each, 42,000 training steps

Metric	Value
Validation reconstruction loss	0.00918
Binary feature accuracy	0.99777
Continuous feature MSE	0.00172
Codebook utilization — books 1/2/3/4	31% / 72% / 81% / 81%
Fixed window w000 binary acc / MSE	1.000 / 0.00174
Fixed window w001 binary acc / MSE	1.000 / 0.00246
Fixed window w003 binary acc / MSE	1.000 / 0.00130

EXP024: S_{plan} is a technically healthy symbolic bottleneck. Books 2–4 reach 72–81% utilization — no codebook collapse. Necessary precondition; does not yet prove downstream acoustic utility.

EXP024: Feature Heatmap — Target vs. Reconstruction

Slakh Track01501, window 0 (0–30 s) — binary acc 1.000, MSE 0.00174



Rows: feature dimensions (binary instrument flags, pitch-class bits, continuous values). Columns: 30 time frames (1 s each). Left panel = ground-truth MIDI features. Right panel = S_{plan} reconstruction.

EXP025: Aligned S_{plan} Reduces Acoustic Token Uncertainty

Proxy prediction head — 5,000 steps, 1,385 test windows

Condition	Valid CE	Test CE	Test PPL
Reason only	7.565	7.657	2116
Low-rate rule summary	7.553	7.645	2089
Aligned S_{plan}	7.507	7.596	1991
$\hookrightarrow$ Shuffled S_{plan}	7.538	7.631	2061
$\hookrightarrow$ Dummy stream	7.602	7.694	2196
$\hookrightarrow$ Random stream	7.540	7.632	2064

Aligned beats reason-only by $\Delta = -0.061$. Aligned beats shuffled by $\Delta = -0.034$ — the alignment-specific effect.

First positive result. Aligned beats both controls. Temporal alignment to the same audio window drives the gain. Claim boundary: proxy task; decoded audio quality is a separate question

Negative Results: Naive Interfaces Do Not Work

EXP027 (side-channel row) and EXP028 (adapter gate)

EXP027 — side-channel row insertion:

Condition	Test CE	vs. reason-only
Reason-only	5.360	baseline
Aligned S_{plan}	5.511	+0.151
Shuffled S_{plan}	5.512	+0.152
Dummy	5.510	+0.150

Aligned $\approx$ shuffled $\approx$ dummy: model ignores symbolic content.

EXP028 adapter: gate converges to 3.4×10^{-4} — training stable, fusion absent.

Lesson: symbolic tokens must enter via the stream-native positional encoding. Side-channel insertion is architecturally incompatible; stability $\neq$ fusion.

EXP029: Stream-Native Interface Produces a Positive Gap

$R_a \rightarrow S_{\text{plan}} \rightarrow C_a$ layout — 3,000 steps, seed 260630

Condition	Test CE	Test Acc	Gate
Aligned S_{plan}	5.766	0.1097	0.003
$\hookrightarrow$ Shuffled	5.785	0.1067	0.003
$\hookrightarrow$ Dummy	5.785	0.1068	0.003

Aligned beats shuffled by $\Delta = -0.018$ and dummy by $\Delta = -0.019$.

Sequence: 151 + 30 + 376 = 557 tokens, **0% truncation**.

Gate (~ 0.003) is small — motivates the longer EXP034B comparison.

Stream-native integration works. Positive gap appears across aggregated test metrics.
Design choice validated over EXP027/028.

EXP034B: Primary Result — Hardened Four-Condition Comparison

All conditions identical except stream variant — 800 steps, `claim_ready = true`

Condition	Test CE	Test PPL	Acc	Gate
Reason-only baseline	5.859	350	0.104	0.000
Aligned S_{plan}	5.600	270	0.121	0.177
$\hookrightarrow$ Shuffled S_{plan}	5.614	274	0.119	0.167
$\hookrightarrow$ Dummy stream	5.614	274	0.119	0.163

Gate confirms the model opens the symbolic slot more for aligned (0.177) than shuffled (0.167) or dummy (0.163).

Strongest result. Aligned beats both corrupted controls ($\Delta = -0.014$ CE) on top of a large capacity gain (-0.245 vs. reason-only).

Boundary: oracle MIDI input; semantic-token CE only; no decoded waveform yet.

EXP034B: Separating Capacity Effect from Alignment Effect

Effect	Δ CE
Stream capacity (any symbolic stream vs. reason-only)	−0.245
Alignment-specific (aligned vs. both controls)	−0.014
Total (aligned vs. reason-only)	−0.259

The alignment-specific gap is consistent across three independent experiments:

Experiment	Setup	Aligned vs. shuffled
EXP025	Proxy head, 5,000 steps	−0.034
EXP029	Stream warmup, 3,000 steps	−0.018
EXP034B	Full hardening, 800 steps	−0.014

Consistent direction across independent experiments rules out a single-run artifact. Capacity effect is dominant; alignment-specific effect is smaller but robust

EXP031A: Music Structure Is Recoverable from S_{plan}

Supervised MIR probing on frozen S_{plan} embeddings — test split, step 625

MIR attribute	Aligned S_{plan}	Zero-feature control
Weak key accuracy	0.557	0.126
Pitch-class macro-F1	0.854	0.830
Instrument macro-F1	0.501	0.426
Segment density acc.	0.874	0.879
Instrument micro-F1	0.791	0.835
Meter accuracy	0.783	0.855

Red = zero-feature control outperforms S_{plan} on that axis.

A zero-feature baseline (head bias only) is mandatory — high accuracy alone is not evidence of symbolic learning. Weak key (+0.431) and pitch-class (+0.024) are the strongest safe axes.

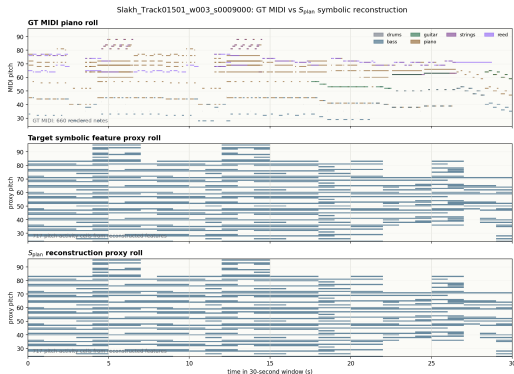
EXP031A: Which Axes Are Safe to Claim

Axis		Gap	Status
Weak key accuracy		+0.431	Safe. Label-shuffle still pending.
Pitch-class	macro-F1	+0.024	Positive but small; target-prior control needed.
Instrument	macro-F1	+0.075	Some signal; class imbalance must be resolved.
Instrument micro-F1		−0.044	Do not headline. Zero-feature wins.
Meter accuracy		−0.072	Do not headline. Common-meter prior dominates.

Paper-safe MIR claim: weak key, pitch-class, instrument macro-F1 — reported alongside the zero-feature gap. not as standalone numbers. Label-shuffle and target-prior controls are

Qualitative: Reconstruction vs. Ground-Truth Piano Roll

Slakh Track01501, window 3 (90–120 s) — binary acc 1.000, MSE 0.00130



Left: ground-truth MIDI piano roll. Right: pitch-region reconstruction from the S_{plan} codec.

Boundary: this is symbolic feature reconstruction — not MIDI transcription or waveform generation.

What We Cannot Claim Yet

Gap	Why it matters
No decoded generation	CE results are on semantic tokens only. No waveform decoded from aligned vs. shuffled.
Oracle MIDI input	Aligned MIDI unavailable at inference on real audio. Needs audio-to- S_{plan} predictor.
No S_{LLM}	Language-supervised tokenizer (open QA, captioning) not yet trained. EXP031B planned.
Pending MIR controls	Label-shuffle and target-prior controls for EXP031A are incomplete.
Slakh only	No real-recording or cross-domain transfer evidence exists.

Next priority: EXP035 decoded rollout — decode C_a from EXP034B aligned checkpoint to waveform and compare perceptually against shuffled.

Conclusion: Evidence Ladder

Exp.	Finding	Status
EXP024	Codec: binary acc 0.998, no codebook collapse	Done
EXP025	Proxy head: aligned CE < shuffled ($\Delta = -0.034$)	Done
EXP027	Side-channel insertion: aligned $\approx$ dummy	Instructive
EXP028	Adapter: stable training but gate ≈ 0	Instructive
EXP029	Stream-native warmup: $\Delta = -0.018$	Done
EXP034B	Hardened 4-cond.: aligned $\Delta = -0.014$ vs. both	Done
EXP031A	Weak key +0.431, pitch-class +0.024 vs. zero-feat.	Done*

*label-shuffle controls pending.

Backup: EXP034B Per-Codebook Cross-Entropy

Aligned S_{plan} achieves the lowest CE in every codebook

Condition	cb0	cb1	cb2	cb3	cb4	cb5	cb6	cb7
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	5.24	5.71	3.19	4.98	5.75	6.25	6.68	7.01
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Symbolic benefit is distributed across all 8 codebooks. Books 0, 2, and 3 show the largest gains, suggesting symbolic structure acts as a long-range prior that helps coarser-grained acoustic prediction first.

Backup: EXP029 Per-Window Case Spectrum

Fixed validation examples — boundary cases included to avoid cherry-picking

Window	Aligned CE	Best control	Margin	Note
w000	5.226	5.280	+0.054	Strong positive
w001	6.159	6.163	+0.004	Small positive
w002	6.097	6.091	−0.006	Boundary / failure
w003	5.876	5.869	−0.007	Boundary / failure

The aggregate positive result (test CE 5.766 vs. 5.785) holds even though individual windows show mixed margins. Including failure cases prevents cherry-picking in qualitative reports.

Backup: EXP030 MIR/Text Target Bank

Data infrastructure for supervised symbolic evaluation

Split	LLM examples	QA examples
Train	89,630	67,072
Validation	18,517	13,849
Test	11,012	8,242
Conversion failures	0 across all splits	

EXP030 links each Slakh window to structured MIR attribute labels and text targets. Enables EXP031A probing now; enables LLM-supervised S_{plan} tokenizer training (EXP031B) after label-shuffle controls pass.